

Nonhomogeneous Equations & Undetermined Coefficients



Complementary function and particular solution

- 1 For $y'' - 3y' + 2y = 4e^{3x}$, find the complementary function y_c .
 - 2 Explain why the general solution of a nonhomogeneous linear equation is $y = y_c + y_p$.
-

Choosing a trial form (no resonance)

- 3 Find a particular solution of $y'' - 3y' + 2y = 4x$.
 - 4 Find a particular solution of $y'' + y' - 2y = 6e^{3x}$ (note $r = 3$ is not a root of $r^2 + r - 2 = 0$).
-

The modification rule (resonance)

- 5 Find a particular solution of $y'' - 3y' + 2y = e^{2x}$. (Note $r = 2$ IS a root of $r^2 - 3r + 2$.)
-

Combining forcing terms

- 6 Find a particular solution of $y'' - y = e^x + x$. (Roots of $r^2 - 1 = 0$ are $r = \pm 1$, so e^x resonates but x does not.)
-

Application: a driven oscillator

- 7 A forced oscillator satisfies $y'' + 4y = \sin 3t$ (forcing frequency $3 \neq$ natural frequency 2 — no resonance). Find a particular solution.
 - 8 Now solve $y'' + 4y = \sin 2t$, where the forcing frequency EQUALS the natural frequency (resonance). Find a particular solution.
-

More practice: no resonance

- 9 Find a particular solution of $y'' - 4y = 8x^2$.
 - 10 Find a particular solution of $y'' + 5y' + 6y = e^x$ (roots of $r^2 + 5r + 6$ are $-2, -3$; no resonance).
-

More practice: with resonance

- 11 Find a particular solution of $y'' - 4y' + 4y = e^{2x}$ (note $r = 2$ is a REPEATED root).

- 12 Find a particular solution of $y'' - y' = 3$ (note $r = 0$ IS a root of $r^2 - r = r(r - 1)$, so constant forcing resonates).
-

Combining a polynomial and an exponential forcing term

- 13 Find a particular solution of $y'' - 3y' + 2y = 2x + e^x$ (roots are 1, 2; e^x resonates, $2x$ does not).
-

Setting up the trial form (without solving)

- 14 For $y'' + 4y' + 3y = 5 \cos 2x$, write the correct trial form for y_p (roots of $r^2 + 4r + 3$ are $-1, -3$; no resonance with sinusoidal forcing).
- 15 For $y'' - 2y' + y = xe^x$ (note $r = 1$ is a REPEATED root, and the forcing already matches that root's exponential), write the correct trial form.
-

Application: a driven oscillator with polynomial forcing

- 16 A forced spring satisfies $y'' + 4y = t^2$ (no resonance, since the roots $\pm 2i$ are not 0). Find a particular solution.
-

One more round of undetermined coefficients

- 17 Find a particular solution of $y'' + y = 3 \cos x$ (roots of $r^2 + 1$ are $\pm i$, so $\cos x$ resonates).
- 18 Find a particular solution of $y'' - 2y' - 3y = e^{4x}$ (roots of $r^2 - 2r - 3$ are 3, -1 ; no resonance).
-

One final trial-form problem

- 19 Find a particular solution of $y'' + 3y' = 9x$ (note $r = 0$ IS a root of $r^2 + 3r = r(r + 3)$, so a bare polynomial resonates).
- 20 Find a particular solution of $y'' + y' - 2y = 10 \sin x$ (roots of $r^2 + r - 2$ are 1, -2 ; no resonance with sinusoidal forcing).
-

A closing pair

- 21 Find a particular solution of $y'' - y = 4$.
- 22 Find a particular solution of $y'' + 4y' + 4y = e^{-2x}$ (note $r = -2$ is a REPEATED root).
-

A final full worked example

23 Solve the initial value problem $y'' - y = 2e^x$, $y(0) = 1$, $y'(0) = 0$ (roots of $r^2 - 1$ are ± 1 , so e^x resonates).